Sky-Guard Radar R3000

3 km Low-Altitude UAV Detection Radar

Overview

The Sky-Guard Radar R3000 is a compact, fully coherent Ku-band pulse-Doppler radar purpose-built for detecting and tracking low-altitude unmanned aerial vehicles. The system comprises a radar unit, a motorised turntable, and display-and-control terminal software. It outputs real-time 3D target data — range, azimuth, altitude, speed, longitude, and latitude — and provides precise cueing to co-located electro-optical or counter-UAS effectors. Maximum detection range is 3 km against small consumer-class UAVs (DJI Phantom 3/4, Pd = 85%).

Operating Mode

360° Full-Domain Low-Altitude Surveillance — the radar continuously sweeps 360° in azimuth while electronically scanning 0°–30° in elevation, automatically initiating track, maintaining target association, and reporting results to the display console.

Applications

- Standalone tripod-mounted deployment for independent UAV detection and tracking
- Integration with EO identification, jamming, or spoofing sub-systems for detect-track-neutralise chains
- Vehicle-mounted (mobile) or rooftop-mounted (fixed) monitoring station configurations

Key Technology

- Fully coherent pulse-Doppler processing
- Sum-and-difference monopulse angle measurement
- MTD coherent integration
- CFAR (Constant False-Alarm Rate) detection

Specifications

Frequency Band	Ku, 16 GHz ± 100 MHz
Azimuth Scan	360° mechanical
Elevation Scan (electronic)	≥ 30°
Beam Width (Az / El)	≤ 3° / ≤ 6°
Peak Transmit Power	≤ 20 W
Max Detection Range	≥ 3 km (DJI Phantom 3/4, Pd = 85%)
Min Detection Range	≤ 250 m
Min Detectable Speed	≤ 2.5 m/s
Range Accuracy (RMS)	≤ 10 m
Azimuth Accuracy (RMS)	≤ 0.5°
Elevation Accuracy (RMS)	≤ 0.5°
Data Refresh Rate	3 s (full 360° sweep)
Radar Unit Dimensions	≤ 650 × 400 × 100 mm
Radar + Turntable Dimensions	≤ 650 × 650 × 250 mm
Power Supply	+28 V DC
Power Consumption	≤ 150 W (incl. turntable)
System Weight	≤ 30 kg (radar + turntable)
Interface	Gigabit Ethernet

Display & Control Software

The accompanying terminal software handles radar operation control, parameter configuration, data parsing, target display, and experimental data logging. It converts native protocols to customer-specified formats for seamless integration with third-party C2 systems.

Packing List

#	Item	Qty	Remark

1	Radar Unit	1	
2	Motorised Turntable	1	
3	Tripod	1	
4	GNSS/INS Antenna	2	
5	Cable Set	1	Power + data, 10 m
6	Fastener Kit	1	

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